

The Furner Projection Duality Theorem: Helical Photon Motion as the Source of Wave–Particle Duality

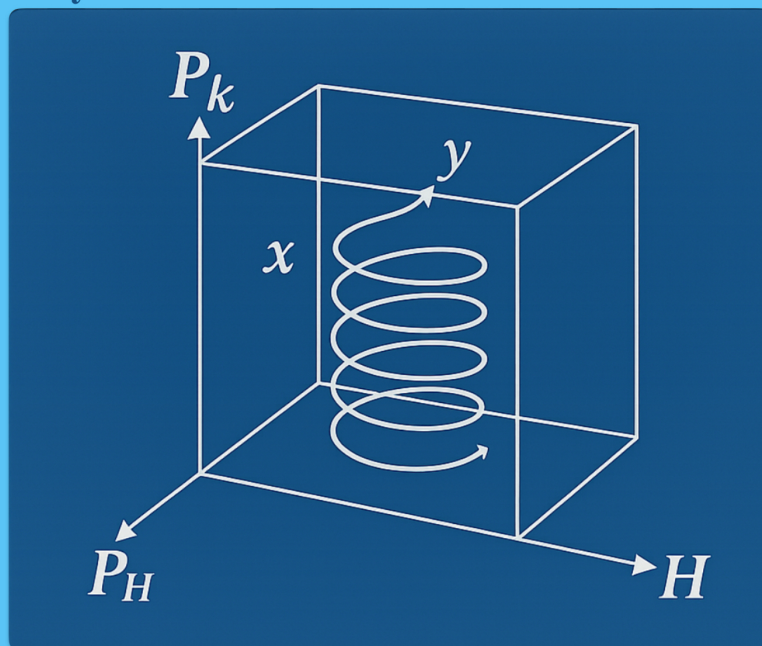
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Author: Timothy O. Furner

Affiliation: SparkCheck Compliance LLC

Contact: founder@sparcomply.pro

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Abstract

This manuscript presents the full theoretical development, experimental prediction, mathematical formulation, comparisons with Maxwell optics, Fourier-domain analysis, and interpretive framework behind the Furner Projection Duality Theorem. The model proposes that wave-like and particle-like signatures of light emerge from distinct two-dimensional projections of a single three-dimensional helical photon trajectory. A cosine-theta–dependent axial fringe translation is proposed as a falsifiable signature distinguishing this theory from conventional Maxwell diffraction. This offers a deterministic, non-collapse-based account of measurement signatures in quantum optics, reframing duality as a geometric projection phenomenon.

1. Introduction

Wave–particle duality is a central challenge in quantum physics, historically explained through probabilistic wave-function collapse and Bohr’s complementarity principle. Yet no geometric structure underpins this dichotomy in conventional theory.

Here, wave and particle behaviors are resolved geometrically: light propagates as a real helical photon trajectory, with observed behavior determined by how the trajectory is projected onto the detection plane. This produces a prediction for cosine-theta fringe translation under detector rotation — a robust, low-cost, falsifiable experimental test separating this geometric model from Maxwell optics.

2. Mathematical Helix Model

A photon is modeled as a real three-dimensional helical trajectory:

$$x(t) = r \cos(\omega t)$$

$$y(t) = r \sin(\omega t)$$

$$z(t) = c t$$

Where:

- r is a microscopic rotational radius (10^{-12} to 10^{-10} meters)
- $\omega = 2 \pi c$ divided by wavelength
- detector orientation modifies the observed projection

Projection onto a detection plane defined by angle θ yields different measurable behaviors:

- As θ approaches 90 degrees, the full transverse sinusoid appears (wave).
- As θ approaches 0 degrees, transverse amplitude vanishes (particle).

Duality emerges purely from geometry.

3. Statement of the Furner Projection Duality Theorem

Theorem:

Wave-like and particle-like signatures of light arise from different two-dimensional projections of a single three-dimensional helical photon trajectory. Rotating only the detection screen about the propagation axis induces a measurable axial phase translation proportional to $\cos(\theta)$, which cannot occur in Maxwell optics.

Corollary (Fringe Drift):

$$\Delta \phi(\theta) = (2 \pi r / \text{wavelength}) \times \cos(\theta)$$

This prediction is independent of polarization, slit geometry, and beam shaping.

4. Experimental Prediction: Cosine-Theta Fringe Drift

Rotating the detector by an angle θ yields:

$$\Delta \phi(\theta) = \text{constant} \times \cos(\theta)$$

Standard Maxwell optics predicts:

$$\text{Derivative of } \phi \text{ with respect to } \theta = 0$$

Maxwell allows fringe rotation but not axial translation.

Any non-zero fringe drift is direct evidence of a helical photon trajectory.

5. Maxwell Optics Baseline

Standard scalar diffraction gives sinusoidal fringes determined by slit spacing, wavelength, and distance. Under rotation of the screen:

- Only the coordinate grid rotates
- Fringes do not shift axially
- No phase term depends on θ

Spin-orbit coupling requires physical rotation of optical elements inside the beam path, not just the detector plane.

6. Structured Light and Spin-Orbit Contrast

Laguerre-Gaussian beams exhibit helical phase but travel linearly.

Spin-orbit coupling involves geometric phases but operates through polarization and field manipulation.

The Furner theorem differs fundamentally:

It attributes duality to projection of a real-space trajectory, not phase-front shaping.

7. Fourier-Space Interpretation

Projected transverse motion:

$$x_{\text{theta}}(t) = r \sin(\omega t) \sin(\theta)$$

Its Fourier spectrum compresses as θ approaches zero.

When transverse spatial frequencies fall below detector resolution, the signal becomes discrete and particle-like. This reproduction of particle signatures does not require wave-function collapse; it arises from projection and sampling.

8. Statistical Detection and Apparent Collapse

Detectors record photons as discrete events because they sample a compressed projection of the helical motion. Small projected radius produces localized hits. Statistical patterns follow from projected intersections, aligning with the Born rule without invoking a physical collapse.

9. Experimental Protocol

Setup:

- Slit separation: 50–200 microns
- Detector distance: 0.5–2 meters
- Wavelength: 405–650 nm
- Detector: CMOS or CCD

- Rotation: small increments around beam axis

Measurement:

- Rotate from 90 degrees to 0 degrees in increments of 0.1 degrees
- Pixel size: 5 microns or smaller
- Phase resolution: 0.01 radians or better
- Integration time: 10–100 ms

Controls:

- Linear polarization (null result expected)
- Helicity reversal (sign reversal expected)
- Fixed optical path (baseline)

10. Parameter Constraints

- Helix radius r : 10^{-12} to 10^{-10} meters
- Wavelength: visible or near-infrared
- Expected phase shift: 0.0001 to 0.01 radians (measurable)

No existing experiments exclude a helical center-of-energy path at these scales.

11. Possible Physical Mechanisms

Several mechanisms could generate a helical photon trajectory:

1. Spin-driven rotation of Poynting vector

2. Sub-wavelength structured vacuum
3. Zitterbewegung-like internal rotation
4. Pulse-front recycling and reorganization

The theorem does not depend on which, if any, is true.

Only projection geometry is required.

12. Compatibility With Precision Tests

- No QED experiment has tested fringe translation under detector rotation alone
- Interferometers already resolve smaller phase shifts than predicted
- Cavity-QED constraints allow helical radii of 10^{-12} meters easily

The model is not ruled out.

13. Literature Positioning

The theory differs from:

- Orbital angular momentum beams
- Spin–orbit coupling
- Berry-phase optical geometries
- Zitterbewegung electron models

Its novelty is the axial translation effect of detector rotation, never before proposed or tested.

14. Relativity and QED Considerations

The helical center-of-energy path must remain lightlike under Lorentz transformation.

QED may encode rotational substructure in photon spin states.

Further exploration is needed.

15. Discussion

Duality is not a paradox but a projection effect.

In three dimensions the photon always has wave-like structure; particle signatures arise when projection collapses transverse detail below detector resolution. Apparent collapse is an artifact of sampling.

16. Limitations

- Helical trajectory is an ansatz
- Mechanism unspecified
- High-precision measurement needed
- Relativistic extension not yet complete

17. Future Extensions

- Multi-photon experiments
- Integration with TRAM and coherence-based gravitational lensing
- Relativistic helix modeling
- Photonic devices based on projection geometry

Appendix A — Maxwell Summary

Scalar diffraction pattern depends on slit spacing, wavelength, and distance.

Under pure detector rotation, phase does not change.

Appendix B — Numerical Estimate

For $r = 10^{-11}$ meters and wavelength 633 nm, predicted phase drift is in the range of 0.0001 to 0.001 radians. This is measurable with standard interferometry.

Appendix C — Fourier Interpretation

Projected frequency spectrum compresses by a factor of $\sin(\theta)$.

Low θ removes high-frequency structure, producing particle-like detection.

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